# Data Analytics project:

## Background:

Over the past few years gas prices have been consistently soaring but its exact causes are unclear. These fluctuations affect daily drivers, businesses and even employment whereby many job candidates must weigh prospective positions against the costs associated with the commute, which they decline if the costs to get to and from work would eat up such a large percentage of the salary ("How Gas Prices Affect the Economy"). While the basic factors like the supply and demand forces and seasonal changes are known to influence gas prices, the recent spikes in prices have an unclear cause. Some speculate that these spikes can be caused by factors like vehicle sales, public transportation usage and crude oil prices (Spiller and Stephens). However, to actually determine if these claims are true, we need to create a model which analyzes past gas price data to actually determine what variables could be used to determine these prices.

## Question:

Can the following independent economic variables be used to estimate gas pump prices? And which combination of these variables creates the best model for it.

Independent variables:

- US Population growth rate (Monthly percentage change in population)
    - Variable name = Pgrowth_rate
    - The proposed explanation for this variable is that an increase in the population will lead to more demand for gas, which will in turn increase the prices of gas, due to demand and supply forces. Therefore, the coefficient of this term should be positive in an ideal model.
- Change in real disposable income in the US (Monthly percentage change)
    - Variable name = Income_rate

    - We suggest that an increase in real disposable income will lead to an increase in demand for gas and therefore a rise in gas prices. The coefficient should be positive in an ideal model.
- Vehicles sold in the US (Millions of units sold)
    - Variable name = Veh_sold
    - The theory for this is that, if the number of vehicles sold increases, then prices will rise, and the coefficient should be positive in an ideal model.
- Public ridership, or the number of separate trips people take in the US (Thousands of unlinked trips)
    - Variable name = Ridership
    - The ridership variable counts the number of public rides taken in the US, which is a substitute to private car ownership. And if people are switching from owning private cars to using public transportation, then prices of gas will fall. Therefore, the coefficient should be negative in an ideal model .
- Global Brent Crude oil prices (Us dollars per barrel)
    - Variable name = Crude_prices
    - The theory behind this is, crude oil prices are the base prices to our gas prices and therefore they should be directly proportional to each other. The coefficient should be positive in an ideal model.
- Federal funds rate (percentage change per month)
    - Variable name = fff_rate
    - This is the base value for interest rates in an economy, therefore a rise in this value will surely decrease demand and therefore gas prices. Meaning an ideal model should have a negative FFF rate coefficient.
- U.S. Dollars to Euro Spot Exchange Rate (Average monthly exchange rate of us dollars for 1 euro)
    - Variable name = us_euro
    - This value tells us how expensive foreign goods are to the US, therefore a higher rate means that foreign goods are more expensive and American purchasing power falls. This would mean that we may demand less gas and prices will fall. Therefore, an ideal model should have this coefficient as negative.

## Null hypothesis:

The null hypothesis of this experiment is that none of the chosen independent variables affect the gas pump prices, meaning the model of this data would have all coefficients equal to zero.

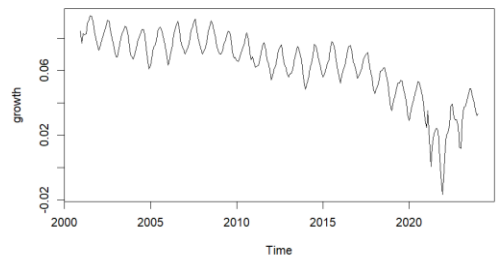
## Alternative Hypothesis:

The alternative hypothesis is that the coefficients of the independent variables have a non-zero value and therefore have a significant correlation to gas pump prices.

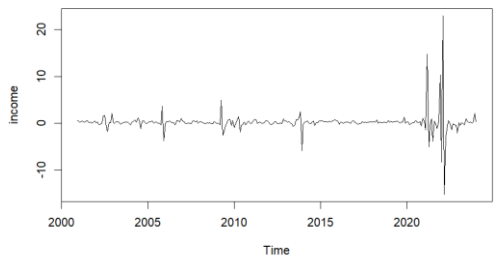
## Data Manipulation:

- To eliminate bias from the data being non-stationary, I used the rate of change in the US population growth and disposable income.
- The data was collected from the FRED website which holds information on all US related economic indicators. The data was collected at a monthly rate from January 2000 to January 2024.
- Next, we change every row of the data into a time series to make it easier for graphing and testing for non-stationarity.

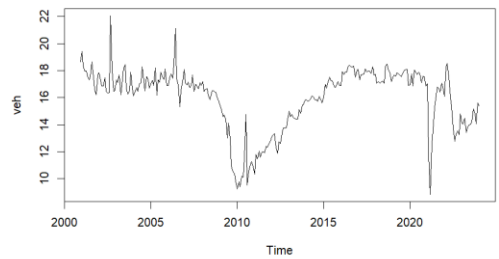
**Plotted graph of variables as Time series**



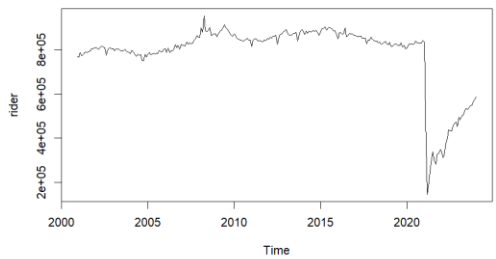

Population growth rate



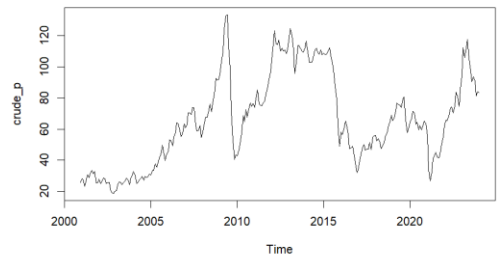

Change in real disposable income



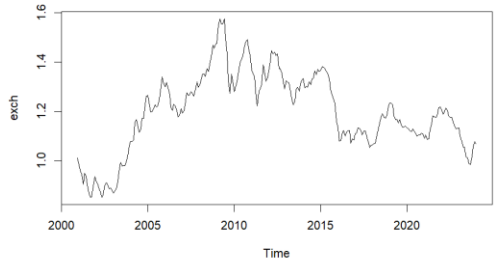

Vehicle sold



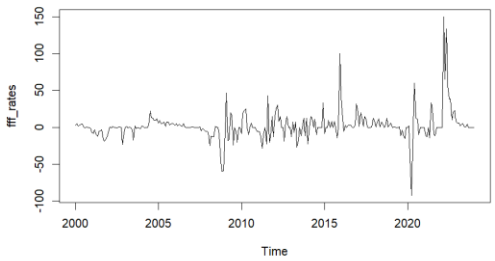

Public ridership



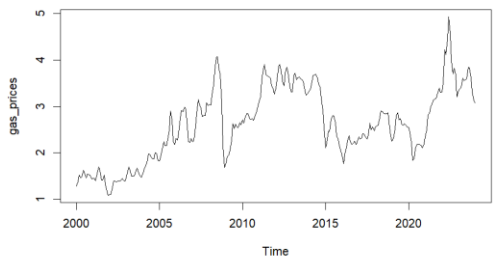

Crude oil prices




Exchange rate




Federal funds rate




Gas prices

- We can notice that most of the data follow a random pattern, and all have some form of outliers between 2020 and 2022, which can be blamed on the covid pandemic which paused the economy and caused these abnormalities in our data.
- We can also tell that none of our data doesn't follow any specific distribution form and therefore we don't need to transform it.
- Looking further at the above plots we notice there are some signs of non-stationary data and to verify this we can use the ADF test.
- The table below shows the p-value of each variable from the ADF test.

| Variable | P-value |
|---|---|
| Population growth | 0.4364 |
| Change in real disposable income | 0.01 |
| Vehicles sold | 0.5075 |
| Public ridership | 0.3847 |
| Crude oil prices | 0.4627 |
| Exchange rate | 0.4339 |
| Change in federal funds rate | 0.01 |
| Gas prices at pump | 0.4313 |

- Looking at the p-values above, we can see that only the population change in real disposable income and the federal funds rate are stationary. Therefore, we need to get the difference of our non-stationary data using the diff() code.

| Variable | P-value |
|---|---|
| Population growth **diff** | 0.01> |
| Change in real disposable income | 0.01> |
| Vehicles sold **diff** | 0.01> |
| Public ridership **diff** | 0.01> |
| Crude oil prices **diff** | 0.01> |

| | |
|---|---|
| Exchange rate **diff** | 0.01> |
| Change in federal funds rate | 0.01> |
| Gas prices at pump **diff** | 0.01> |

- The table of our new differentiated data shows that all the p-values are less than 0.01 and we can therefore confidently say that all our data is non-stationary.
- Now we can create a model for this data. A model with all the variables used to predict gas prices looks like this.

```
##
## Call:
## lm(formula = diff_gas ~ ., data = gas_m)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.49242 -0.06220 -0.00076  0.07099  0.40494
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)    1.278e-03  6.613e-03   0.193  0.84688
## diff_growth    7.994e+00  1.344e+00   5.946 8.16e-09 ***
## diff_vehicles -8.333e-03  7.229e-03  -1.153  0.25002
## diff_crude     2.160e-02  1.262e-03  17.119  < 2e-16 ***
## diff_exch     -3.259e-02  2.591e-01  -0.126  0.90000
## change_inc     9.559e-04  3.144e-03   0.304  0.76134
## fffr           9.150e-04  3.511e-04   2.606  0.00966 **
## diff_rider     1.439e-08  2.340e-07   0.061  0.95102
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1107 on 280 degrees of freedom
## Multiple R-squared:  0.6133, Adjusted R-squared:  0.6037
## F-statistic: 63.45 on 7 and 280 DF,  p-value: < 2.2e-16
```

- Analyzing the output, we can see that the model is successful at explaining a lot of the changes in gas pump prices as the R-Squared value is 0.6, but only three of our coefficients are significant.
- If we create another model with just our significant coefficients, we can get a better a more accurate and efficient model.

```
Call:
## lm(formula = diff_gas ~ diff_growth + diff_crude + fffr)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.49434 -0.06194  0.00029  0.06653  0.40856
##
## Coefficients:
```

```
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 0.0016872  0.0065393   0.258   0.7966
## diff_growth 8.0585355  1.3363879   6.030 5.09e-09 ***
## diff_crude  0.0212847  0.0011519  18.478  < 2e-16 ***
## fffr        0.0008857  0.0003352   2.642   0.0087 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1102 on 284 degrees of freedom
## Multiple R-squared:  0.6112, Adjusted R-squared:  0.6071
## F-statistic: 148.8 on 3 and 284 DF,  p-value: < 2.2e-16
```

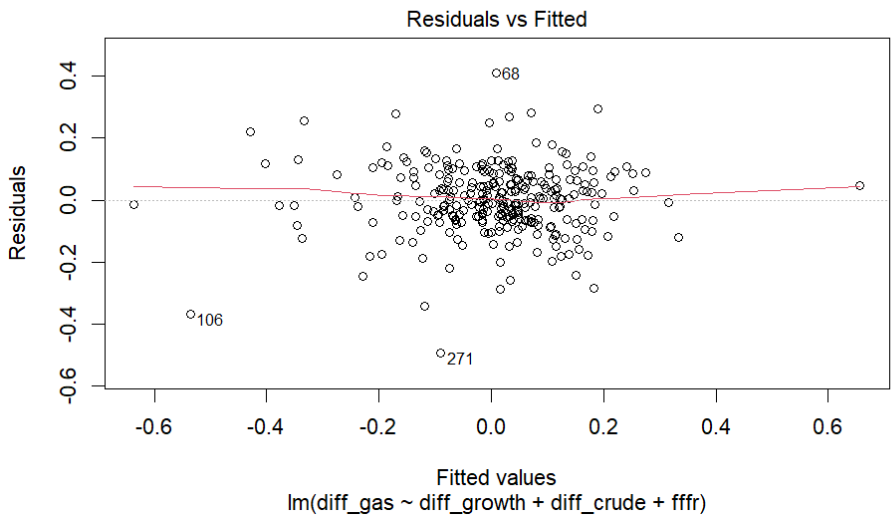
- This new model shows an improvement in the accuracy of the model as the adjusted R-squared value is slightly larger.
- By analyzing this new model, we can see that the change in the rate of population growth has the largest effect on these gas prices, as an increase in the rate of population growth rate by 1 will cause a rise in rate of increase in gas prices per gallon by about 8.
- The federal funds rate on the other hand has the lowest effect, as an increase in the rate by 1 only causes a rise in the rate of increase in gas prices per gallon by about 0.0009.
- Crude oil prices seem to be the best at predicting gas prices as it has the lowest p-value, but its effect on the predicted value of gas price rates is minimal. Whereby a rise in the rate of increase of crude oil prices by 1 causes a rise in the rate of increase in gas prices per gallon by about 0.02.
- Next, we must compare these coefficients to those of the ideal model and evaluate the model to check for violations of the linear model assumptions.

## Hypothesis comparison:

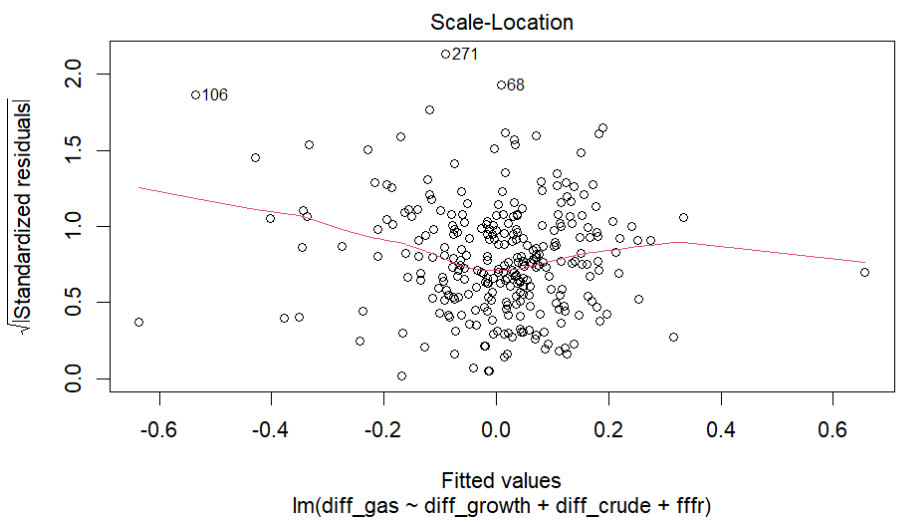
- Comparing the first model to the ideal model predictions we notice that the predictions for the federal funds rate, ridership and vehicle sales differ from the actual coefficients obtained from the model. But overall, both models show that the selected independent variables have some form of correlation to the gas prices, therefore we reject the null hypothesis.

## Residuals:

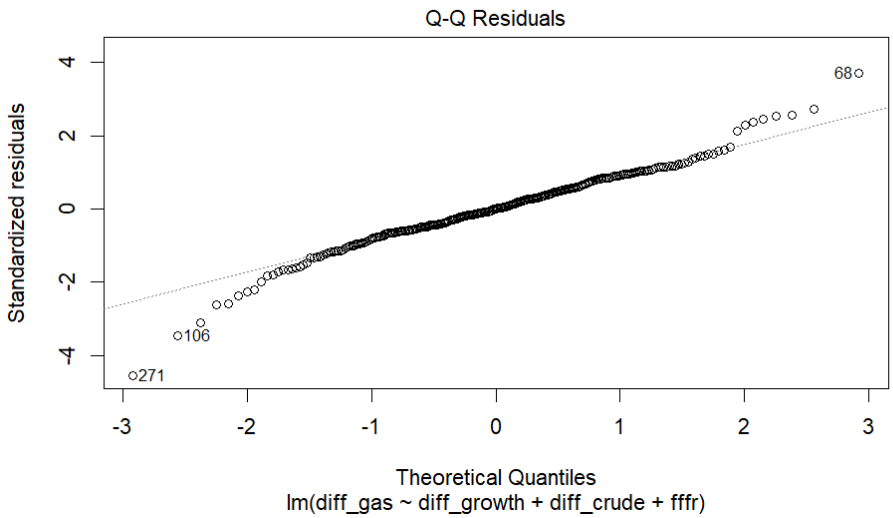
- Now that we have created a model, we need to check to see if it follows the assumptions of a linear model using residual plots.



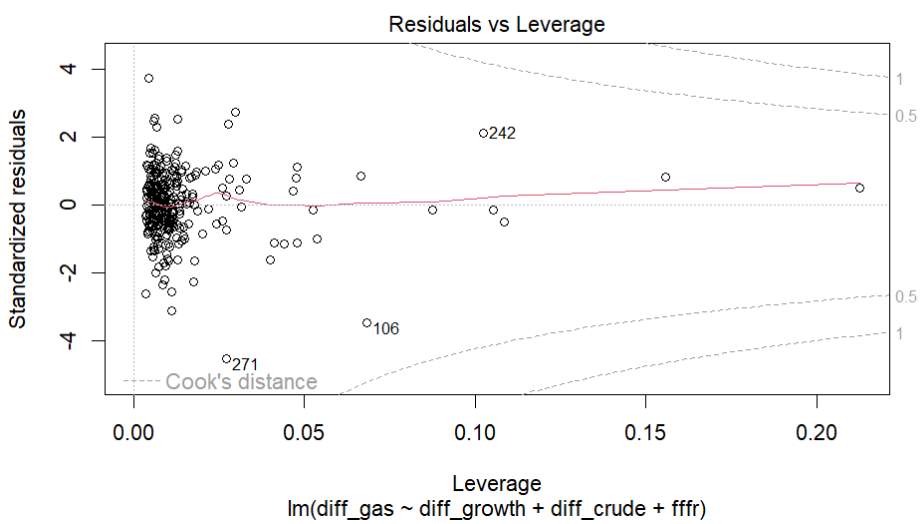

- Looking at the residual vs Fitted plot we can tell that the model doesn't violate these assumptions as the values are fairly distributed below and above the dotted line.




- Whereas on the other hand the scale-location plot shows that the is a fall in variation as the number of fitted values increase, but this fall isn't significant, and our model still conforms to the linear assumptions.




- Overall, the Q-Q residual plot suggests that there model has a little more variation than the normal linear model but isn't a significant increase, as most of the values in the plot go along the normal line.




- The residual vs leverage plot also suggests that our model conforms with the linear model assumptions, as little values are so extreme that it affects our predictions.
- Therefore, we can conclude that our model isn't problematic.

## Works Cited

"How Gas Prices Affect the Economy." *Investopedia*, www.investopedia.com/financial-edge/0511/how-gas-prices-affect-the-economy.aspx#:~:text=ability%20to%20hire.-. Accessed 2 May 2024.

Spiller, Beia, and Heather Stephens. "Gas Prices 101." *Resources for the Future*, 9 Sept. 2022, www.rff.org/publications/explainers/gas-prices-101/.